



Calculation Report 0040495517_001[00]

Subject:	SiL Model for JLR MLA High SCU4D6 L461	
Change Index:	[00]	
Handled by:	Krishna Kanth Jayasimman (MS/ESY8-VM)	CAE Task: CAE023231
Client:	Soundariya B (MS/ESY2-VM)	Part No.: see 3.1
Date:	13th August, 2025	Task Def.: -
		Base report: -
Circulation:	Ram Prakash R S(MS/ESY8-VM), Noupa Karl Max Crepin(VM-EPS/ENC4), Soundariya B (MS/ESY2-VM), Baallaoui Abdelmjid (VM-EPS/ENC1), Ilg Michael, Kohlmann Christian (VM/EPS-ENC4), Krishna Kanth Jayasimman (MS/ESY8-VM)	

1 Task

Provide SiL models according to technical specification in 3.2 for JLR MLA High LHD & RHD based on the design specified in 3.1.

2 Results

The SiL models are available. From VM/ENC4's point of view, the models can be authorized and submitted to the customer/OEM.

CAE internal link	CAE Internal Link
SiL Release View	-
vHW Library	Stubbed and full HWlib included. vHWlib: H2E_AEI_A_10_00_C16/H2E_AEJ_A_10_00_C16
Report of the vEPS - Parameter Identification and Comparison to Measurement	-
Final recipient	JLR

3 Conclusion

According to the current procedure on model exchange, which is described in [AS Methods Library](#), models created by VM/ENC4 have to be approved and forwarded to customer by client (e.g., SYS-PL). If these models cannot be approved, an iteration loop involving VM/ENC4 is necessary.



VM/ENC4 will support client if she/ he is not sure whether SiL models can be approved or not and for transferring the model to the final customer/OEM.

Good to know: SiL models enable Bosch AS to do further investigations (see attachment 3.3) either on steering system or on vehicle level. Several investigations require input data representing the dynamic behaviour of vehicle components (e.g., vehicle axis). Those input data could be provided by AS/ENS. Kindly come up to the corresponding department for further information.

8/26/2025
X
Krishna Kanth Jayasimman
Author
Signed by: Jayasimman_KrishnaKanth

26.08.2025
X
Baallaoui Abdelmjid
CAE Project Coordinator
Signiert von: Abdelmjid_Baallaoui

8/26/2025
X
Ram Prakash Rengasamy Sethuraman
Manager MS/ESY8-VM
Signed by: RS_RamPrakash

26.08.2025
X
Noupa Karl Max Crepin
L5 expert model assembly
Signiert von: KarlMaxCrepin_Noupa



Calculation Report 0040495517_001[00]

3.1 Gear Variants

Gear Name	Software release [OEM BAS naming]	Steering gear part number
EPSapa LHD iconst	JLR 450_700_C_05_00	4532.051.963-01
EPSapa RHD iconst	JLR 450_700_C_05_00	4532.051.964-01

3.2 Technical Specification

Customer's requirements to the virtual EPS are defined in the link below:

- [DOORS](#)

3.3 Use Cases applying SiL models

SiL Models [vECU | vEPS] as enabler for various use cases

● vEPS Level ● vECU Level

Use Case	Description	Alternative	Importance / Saving potential
● Safe by Design (HAD)	▪ Required for L4 release for HAD systems	N/A	Blocker for HAD
● Vehicle Motion Control (VMC)	▪ Required for VMC release for JLR at Bosch CC	N/A	Blocker for VMC for Bosch CC
● Component Integration Test (CiT)	▪ Required for all software releases	N/A	
● HiL2SiL	▪ Test case generation via SiL ▪ Parallel implementation of test cases	HiL	Avoid necessity of additional HiL's (0,5 Mio € over all projects)
● Model Exchange OEM	▪ Fulfill OEM requirements	N/A	Avoid potential payback of dev. costs
● Virtualized Steering Stability Process (VSSP)	▪ Objective evaluation of system stability ▪ Partially OEM requirement (Toyota, Ford etc.) ▪ Required for L3 release (pklt v5.x)	N/A	Avoid late detection of issues → Field problems. Decreased efforts in vehicle tests, increased system test coverage
● Virtualized Abnormal Noise Process (VANP)	▪ Objective evaluation of rattle & clunk potential	N/A	Avoid late detection of issues → Field problems. Reduce costs for prototypes with bad design regarding rattle & clunk
● Virtual Test Bench	▪ Virtualized SCU for functional testing and debugging	Test bench with SCU	Reduce number of SCU samples, Alternative for "Lenkungswagen"
● Analysis of Dynamic Load Cases (ALD)	▪ Evaluation of component loads for the dimensioning of the Rack, Pinion, BNG and Housing.	N/A	Late detection of issues.
● Virtualized Shimmy Compensation Tuning (VSCT)	▪ Identification of SW parameters for Shimmy Compensation based on EPS transfer behavior.	Vehicle or bench test	Additional cost for vehicle or bench tests.



Calculation Report 0040495517_001[00]

3.4 Parameter

3.4.1 Model-Info

Following model parameters for steering gear 4532_051_963_01 were provided by Soundariya, B (MS/ESY2-VM) and derived from existing toothing data.

Information	
Customer	JLR
Project	MLA_High_SCU4_D6a
Number of list of parts (steering gear)	4532_051_963_01
Software Label	450_700_C_05_00
Motor	M82S
Manufacturer	Nidec
Dimension	
Max. Rack Travel [mm]	95.7
Rack Diameter [mm]	28
Max. Parking Force [N]	14500
Max. Servo Force [N]	14154
Mass	
Rack [kg]	3.242
KGT Nut [kg]	1.257
Outer Bearing Ring [kg]	0.361
Mass Moment of Inertia	
Motor (shaft with rotor) [kgmm ²]	133.1
Tooth Wheel at Engine Shaft [kgmm ²]	44.76
KGT Package [kgmm ²]	951.94
Steering Pinion [kgmm ²]	25.67
Input Shaft + Sensor [kgmm ²]	14.71
Gear Ratio	
Toothing Ratio [mm/rev]	79.9
Toothing Number	V17320
Belt [-]	2.4894
KGT pitch [mm/rev]	7
Torsion Bar	
Torsion bar stiffness [Nm/°]	2.5
Max. Torsion bar deflection [deg]	4.8
Design KGT	
KGT Thread Generation	3
Outer BNG spring	WF_243
Installation space outer BNG spring [mm]	0.2
Inner BNG spring	WF_224
Installation space inner wave spring [mm]	0.023
Bearing Type	Double Row / Pressure Angle 40[deg]
Friction	
Friction Motor [Ncm]	2.755
Friction NUT [Ncm]	12
Yoke spring force [N]	260
Yoke Contact angle [°]	45

Fulfils requirement /
 Unclear if requirement fulfilled /
 Violates requirement



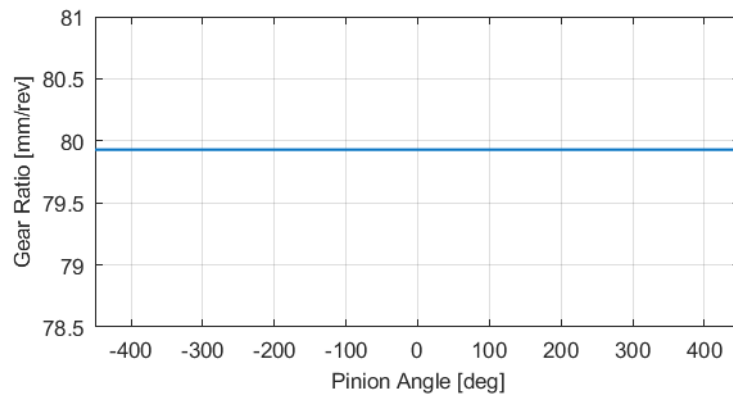
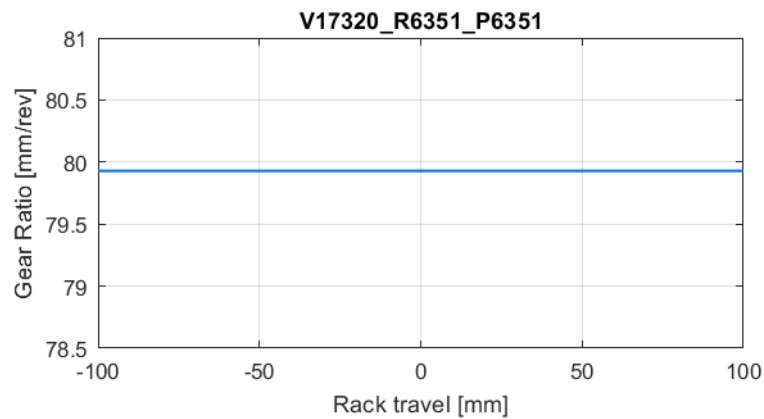
Calculation Report 0040495517_001[00]

Rack-Pin-Gear Transmission

The following rack-pin-gear toothing data was extracted from technical report RS146909.

ITE calculation name:

Sensor Toothing	Values
rack number	R6351
pinion number	P6351
VERA calculation	V17320
ratio [mm/rev]	79.9





Calculation Report 0040495517_001[00]

3.4.2 Model-Info

Following model parameters for steering gear 4532_051_964_01 were provided by Soundariya, B (MS/ESY2-VM) and derived from existing toothing data.

Information	
Customer	JLR
Project	MLA_High_SCU4_D6a
Number of list of parts (steering gear)	4532_051_964_01
Software Label	450_700_C_05_00
Motor	M82S
Manufacturer	Nidec
Dimension	
Max. Rack Travel [mm]	95.7
Rack Diameter [mm]	28
Max. Parking Force [N]	14500
Max. Servo Force [N]	14154
Mass	
Rack [kg]	3.242
KGT Nut [kg]	1.257
Outer Bearing Ring [kg]	0.361
Mass Moment of Inertia	
Motor (shaft with rotor) [kgmm ²]	133.1
Tooth Wheel at Engine Shaft [kgmm ²]	44.76
KGT Package [kgmm ²]	951.94
Steering Pinion [kgmm ²]	25.67
Input Shaft + Sensor [kgmm ²]	14.71
Gear Ratio	
Toothing Ratio [mm/rev]	79.9
Toothing Number	V17214
Belt [-]	2.4894
KGT pitch [mm/rev]	7
Torsion Bar	
Torsion bar stiffness [Nm/°]	2.5
Max. Torsion bar deflection [deg]	4.8
Design KGT	
KGT Thread Generation	3
Outer BNG spring	WF_243
Installation space outer BNG spring [mm]	0.2
Inner BNG spring	WF_224
Installation space inner wave spring [mm]	0.023
Bearing Type	Double Row / Pressure Angle 40[deg]
Friction	
Friction Motor [Ncm]	2.755
Friction NUT [Ncm]	12
Yoke spring force [N]	260
Yoke Contact angle [°]	45



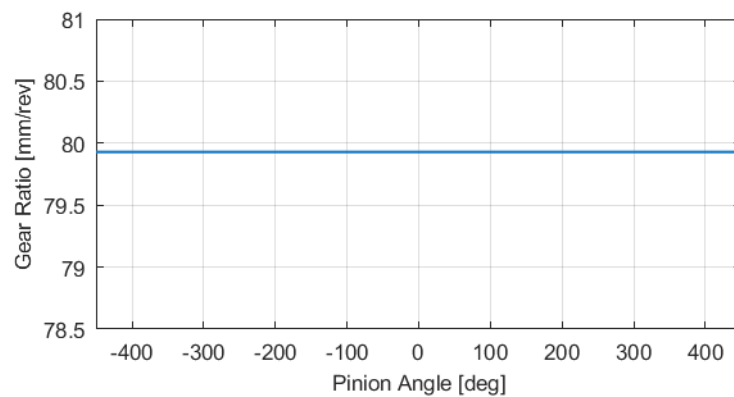
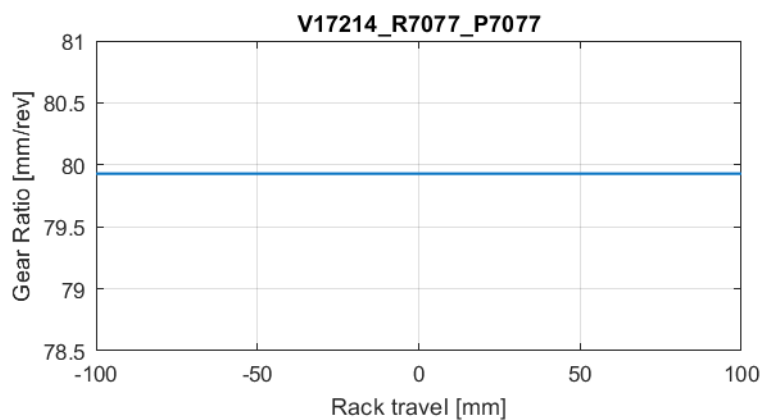
Calculation Report 0040495517_001[00]

Rack-Pin-Gear Transmission

The following rack-pin-gear toothing data was extracted from technical report RS146654.

ITE calculation name:

Sensor Toothing	Values
rack number	R7077
pinion number	P7077
VERA calculation	V17214
ratio [mm/rev]	79.9



3.5 Release Checklist

All following points must be checked to release the vEPS. A checked point below means, it was checked and implemented correctly.

3.5.1 EPS_load_param_set

Check	Item
☒	Content is adapted to the specific release content of the OEM / project. (no Content of different projects / OEMs)
☒	Only valid and used features are available
☒	All not used features / parts are deleted (not commented)

☒ ☐ ☐ Fulfils requirement /
 ☐ ☐ ☐ Unclear if requirement fulfilled /
 ☐ ☐ ☒ Violates requirement



Calculation Report 0040495517_001[00]

☒	help comment is updated for the current release / OEM / project including valid examples
☒	Function must run with default settings (without Parameters)
☒	All selectable parameters can be changed via function call (e.g. EPS_load_param_set('Param_1', <VALUE>);)
☒	All valid parameter files and model configurations must be selectable

3.5.2 Modell Library

Check Item

☒	All external Links are broken
☒	The Model Library contains all needed Blocks (Submodels / Components) as well as example model(s)
☒	The BoschAS SteeringSystem Library was checked for updates
☒	Example models are only using native Simulink Blocks and Linked Blocks from the Components

3.5.3 Documentation

Check Item

☒	All internal links in the documentation are checked and valid, especially the links in the navigation bar e.g. Release_Notes
☒	Parameter and configuration of the function EPS_load_param_set are explained including examples how to select them.
☒	The files page was automatically updated with the corresponding python file
☒	System Requirements are updated
☒	The Main page is updated with the current release information
☒	The vECU interface must be described • BusIO • Configurations • Parameter (xml files)
☒	No unused / invalid folders of the documentation are available.

3.5.4 Content model exchange

Check Item

☒	Common Folder Structure is used
☒	All folders are updated with the correct content and empty folders are deleted
☒	All required tools are available in the exchange folder e.g.: • Sofia • ZFLSparfile_change • Readpar mexfunction • ...
☒	Check if validated parameter file is available (From "Sil Model Validation" task) No, not available.
☒	Check if Linux or Adams upgrade tasks are available.



Calculation Report 0040495517_001[00]

	Yes, Linux task is available and task status is updated
--	---

3.5.5 Model Test

Check	Item
☒	Example model must run in customer environment without any adoptions
☒	Are all (automized) release tests are successful